

Serverless Architectures: Demystifying Serverless Computing

Serverless architectures refer to applications that depend a lot on third party services known as BaaS (Backend as a Service), or on custom code which runs on FaaS (Function as a Service).



In the 1990s, Neal Ford (now at ThoughtWorks) was working in a small company that focused on a technology called Clipper. By writing an object-oriented framework based on Clipper, DOS applications were built using dBase. With the expertise the firm had on Clipper, it ran a thriving training and consulting business. Then, all of a sudden, this Clipper-based business disappeared with the rise of Windows. So Neal Ford and his team went scrambling to learn and adopt new technologies. “Ignore the march of technology at your peril,” is the lesson that one can learn from this experience.

Many of us live inside ‘technology bubbles’. It is easy to get cozy and lose track of what is happening around us. All of a sudden, when the bubble bursts, we are left scrambling to find a new job or business. Hence, it is important to stay relevant. In the 90s, that meant catching up with things like graphical user interfaces (GUIs), client/server technologies and later, the World Wide Web. Today, relevance is all about being agile and leveraging the cloud, machine learning, artificial intelligence, etc.

With this background, let’s delve into serverless computing, which is an emerging field. In this article, readers will learn how to employ the serverless approach in their applications and discover key serverless technologies; we

will end the discussion by looking at the limitations of the serverless approach.

Why serverless?

Most of us remember using server machines of one form or another. We remember logging remotely to server machines and working with them for hours. We had cute names for the servers - Bailey, Daisy, Charlie, Ginger, and Teddy - treating them well and taking care of them fondly. However, there were many problems in using physical servers like these:

- Companies had to do capacity planning and predict their future resource requirements.
- Purchasing servers meant high capital expenses (capex) for companies.
- We had to follow lengthy procurement processes to purchase new servers.
- We had to patch and maintain the servers ... and so on.

The cloud and virtualisation provided a level of flexibility that we hadn’t known with physical servers. We didn’t have to follow lengthy procurement processes, or worry about who ‘owns the server’, or why only a particular team had ‘exclusive access to that powerful server’, etc. The task of procuring physical machines became obsolete with the arrival